SECTION 01 1100 – SUMMARY OF WORK

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract and associated addenda, apply to this section.

1.2 SUMMARY

A. This Section includes the following:

1. Work covered by the Contract Documents.
2. Type of the Contract.
3. Work phases.
4. Work under other contracts.
5. Use of premises.
6. Owner's occupancy requirements.
7. Work restrictions.
8. Specification formats and conventions.

1.3 WORK COVERED BY CONTRACT DOCUMENTS

A. Project Identification: Johnson Dean Preserve

1. Project Location: 145 North Barton St, Norcross, GA 30071

B. Owner: The City of Norcross (65 Lawrenceville Street, Norcross, GA 30071)

C. Landscape Architect of Record: Root Design Studio, 2300 Henderson Mill Rd, Suite 412, Atlanta, GA 30345.

D. The Work consists of the following:

1. The Work includes but is not limited to the construction of a new driveway, pavilion/restroom building, parking lot, concrete paving, drainage improvements, utility improvements, retaining wall, fencing, electrical improvements, bioswales and ornamental landscaping.

E. Project will be constructed under a general construction contract.

1.4 WORK PHASES

A. The Work shall be conducted in one phase:

1. Phase One: The Work includes but is not limited to the construction of a new driveway, pavilion/restroom building, parking lot, concrete paving, drainage improvements, utility improvements, retaining wall, fencing, electrical improvements, and ornamental landscaping.

B. Before commencing Work, submit a schedule showing the sequence, commencement and completion dates, for all aspects of the Work.

C. A preconstruction conference between, at a minimum, the Owner, contractor, and the design professional(s) shall be conducted prior to commencement of work.

1.5 USE OF PREMISES

A. General: Contractor shall have full use of premises for construction operations, including use of Project site, during construction period. Contractor's use of premises is limited only by Owner's right to perform work or to retain other contractors on portions of Project.

1.6 WORK RESTRICTIONS

A. Existing Utility Interruptions: Do not interrupt utilities serving facilities occupied by Owner or others unless permitted under the following conditions and then only after arranging to provide temporary utility services according to requirements indicated:

1. Notify Owner not less than two days in advance of proposed utility interruptions.

2. Do not proceed with utility interruptions without Owner’s written permission.

1.7 SPECIFICATION FORMATS AND CONVENTIONS

A. Specification Format: The Specifications are organized into Divisions and Sections using the 50-division format and CSI/CSC's "MasterFormat" numbering system.

1. Section Identification: The Specifications use Section numbers and titles to help cross-referencing in the Contract Documents. Sections in the Project Manual are in numeric sequence; however, the sequence is incomplete because all available Section numbers are not used. Consult the table of contents at the beginning of the Project Manual to determine numbers and names of Sections in the Contract Documents.

2. Division 01: Sections in Division 01 govern the execution of the Work of all Sections in the Specifications.

B. Specification Content: The Specifications use certain conventions for the style of language and the intended meaning of certain terms, words, and phrases when used in particular situations. These conventions are as follows:

1. Abbreviated Language: Language used in the Specifications and other Contract Documents is abbreviated. Words and meanings shall be interpreted as appropriate. Words implied, but not stated, shall be inferred as the sense requires. Singular words shall be interpreted as plural, and plural words shall be interpreted as singular where applicable as the context of the Contract Documents indicates.

2. Imperative mood and streamlined language are generally used in the Specifications. Requirements expressed in the imperative mood are to be performed by Contractor. Occasionally, the indicative or subjunctive mood may be used in the Section Text for clarity to describe responsibilities that must be fulfilled indirectly by Contractor or by others when so noted.

a. The words "shall," "shall be," or "shall comply with," depending on the context, are implied where a colon (:) is used within a sentence or phrase.

1.8 ONE YEAR WARRANTY INSPECTION

A. The Owner and Contractor will schedule an inspection within 10 days of the one-year anniversary of the date of Final Acceptance. Any Work found not to be in accordance with the Contract Documents will be corrected on a schedule agreed upon by all parties. Work shall be corrected at the Contractor’s expense.

END OF SECTION 01 1100

SECTION 01 2500 – SUBSTITUTIONS

PART 1 GENERAL

1.01 SECTION INCLUDES:

A. Requirements included

B. Substitutions / prior approvals

C. Submittal requirements

D. Contractor's representation

1.02 REQUIREMENTS INCLUDED:

A. Substitutions for products specified shall be allowed only under the conditions stated in this section.

1.03 SUBSTITUTIONS/PRIOR APPROVALS:

A. If it is desired to use products different from those indicated in the Contract documents, the party requesting the substitution shall make written application as described herein. The burden of proving equality of proposed substitutions rests on the party making the request for substitution.

B. The Owner is under no obligation to consider substitution requests if they are received less than ten (10) working days prior to the bid opening nor if they are submitted by the awarded contractor after the bid opening and anytime during construction.

PART 2 PRODUCTS (NOT USED)

PART 3 EXECUTION

3.1 SUBMITTAL REQUIREMENTS:

A. Submit a separate request for each substitution.

B. Support each request with the following information:

1. Date of request

2. Name of party proposing substitution.

3. Project name.

4. Specification reference.

5. Complete data substantiating compliance of proposed substitution with requirements stated in Contract Documents:

    i. Product identification, including manufacturer's name and address.

    ii. Manufacturer's literature, identify:

        a. Product description.

        b. Reference standards.

        c. Performance and test data.

        d. Manufacturer's recommendation for use and installation.

    iii. Samples, as applicable.

    iv. Name and address of similar projects on which product has been used, and date of each installation.

6. Itemized comparison of the proposed substitution with product specified, list all variations.

7. Data relating to changes in construction schedule.

8. Any effect of substitution on separate contracts.

9. List of changes required in other work or products.

10. Designation of required license fees or royalties.

11. Designation of availability of maintenance services and sources of replacement materials.

3.2 CONTRACTOR'S REPRESENTATION

A. In connection with the use of any substitute item approved by the Design Professional(s), it shall be the General Contractor's responsibility to see that such items meet all space requirements, and that any alterations to connecting items necessitated by use of the alternate items are properly made at no increase in cost to the Owner, and that all items are in compliance with the specification requirements. Contractor shall waive all

claims for additional costs caused by substitution which may subsequently become apparent.

END OF SECTION 01 2500

SECTION 01 2600 – CONTRACT MODIFICATION PROCEDURES

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of Contract including General and Supplementary Conditions and other Division 1 Specifications Sections, apply to this section.

1.02 SUMMARY

A. This section specifies administrative and procedural requirements for handling and processing Contract modifications.

1.03 MINOR CHANGES IN WORK

A. Supplemental instructions authorizing minor changes in the work, not involving an adjustment to the Contract Sum or Contract Time, will be issued by the Design Professional(s) in the form of a memorandum or Field Order.

1.04 CHANGE ORDER PROPOSAL REQUESTS

A. Owner-Initiated Proposal Requests: Proposed changes in the work that will require adjustment to the Contract Sum or Contract Time will be issued by the Design Professional(s), with a detailed description of the proposed change, and supplemental or revised Drawings and Specifications, if necessary.

1. Proposal requests issued by the Design Professional(s) are for information only. Do not consider them an instruction either to stop work in progress, or to execute the proposed change.

2. Unless otherwise indicated in the proposal request, within 10 days of receipt of the proposal request, submit to the Design Professional(s) for the Owner's review an estimate of cost necessary to execute the proposed change.

B. Contractor-Initiated Change Order Proposal Requests: When latent or other unforeseen conditions require modifications to the Contract, the Contractor may propose changes by submitting a request for a change to the Design Professional(s).

1. Include a statement outlining the reason for making the change and the effect of the change on the work. Provide a complete description

of the change. Indicate the effect of the proposed change on the Contract Sum and Contract Time.

C. Proposal Request Form: Change Order Proposal Requests will be documented by memorandum.

1.05 CHANGE ORDER PROCEDURES

A. Upon the Owner's approval of a Change Order Proposal Request; the Design Professional(s) will issue a Change Order for signatures of the Owner, and Contractor on AIA Form G 701 or other approved form, as provided in the Conditions of the Contract.

END OF SECTION 01 2600

SECTION 01 2900 – PAYMENT PROCEDURES

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this section.

1.02 SUMMARY

A. This Section specifies administrative and procedural requirements governing the Contractor's Applications for Payment.

1.03 APPLICATIONS FOR PAYMENT

A. Each Application for Payment shall be consistent with previous applications and payments as certified by the Design Professional(s) and paid for by the Owner.

B. Payment Application Times: Each progress payment date is as indicated in the Agreement. The period of construction work covered by each Application for Payment is the period indicated in the Agreement.

C. Payment Application Forms: Use AIA Document G 702 and Continuation Sheets G 703 as the form for Application for Payment unless otherwise approved by the Owner.

D. Application Preparation: Complete every entry on the form, including notarization and execution by person authorized to sign legal documents on behalf of the Owner. Incomplete Applications will be returned without action.

1. Entries shall match data on the Schedule of Values and Contractor's Construction Schedule. Use updated schedules if revisions have been made.

2. Include amounts of Change Orders issued prior to the last day of the construction period covered by the application.

E. Transmittal: Submit 3 executed copies of each Application for Payment to the Owner at each monthly progress meeting. All copies shall be complete, including waivers of lien when required.

F. Release of Lien Waivers: With final application for Payment, submit waivers from every entity (including Contractor) who may lawfully be entitled to file a lien arising out of the Contract, and related to the work covered by the Payment.

1. Waiver Forms: Submit waiver on forms, and executed in a manner, as specified; utilize AIA Document G 706 A.

F. Initial Application for Payment: Administrative actions and submittals that must precede or coincide with submittal of the first Application for Payment include the following:

1. List of Subcontractors and principal suppliers and fabricators.

2. Contractors construction schedule.

3. Schedule of values.

4. Copies of building permits.

5. Certificates of insurance.

6. Schedule of submittals.

END OF SECTION 01 2900

SECTION 01 3300 – SUBMITTAL PROCEDURES

PART 1 GENERAL

1.01 RELATED DOCUMENTS

A. Drawings and general provisions of Contract, including General and Supplementary Conditions and other Division 1 Specification Sections, apply to this section.

1.02 SUMMARY

A. This section specifies administrative and procedural requirements for submittals required for performance of the work, including:

1. Contractor's construction schedule.
2. Submittal schedule.
3. Shop Drawings.
4. Product Data.
5. Samples.
6. Miscellaneous work - related submittals.

B. Administrative Submittals: Refer to other Division 1 Sections and other Contract Documents for requirements for administrative submittals.

1.03 SUBMITTAL PROCEDURES

A. Coordination: Coordinate preparation and processing of submittals with performance of construction activities. Transmit each submittal sufficiently in advance of performance of related construction activities to avoid delay.

B. Processing: Allow sufficient review time so that installation will not be delayed as a result of the time required to process submittals, including time for resubmittals.

C. Submittal Preparation: Place a permanent label or title block on each submittal for identification. Indicate the name of the entity that prepared each submittal on the label or title block. Include the following information on the label for processing and recording action taken

1. Project Name.

2. Date.

3. Name and address of contractor, sub-contractor and/or supplier.

4. Name of manufacturer.

5. Name of product, model number, use and location with Specification Section number.

6. Field dimensions; clearly identified as such.

7. Applicable standards, such as ASTM or Federal Specification numbers.

8. Drawing number and detail references, as appropriate.

9. Contractors stamp, initialed or signed and dated.

D. Submittal Transmittal: Package each submittal appropriately for transmittal and handling. Transmit each submittal from Contractor to the Design Professional(s) using a transmittal form.

1. On the transmittal record relevant information and requests for data. On the form, or separate sheet, record deviations from Contract Document requirements, including minor variations and limitations. Include Contractor's certification that information complies with Contract Document requirements.

1.04 CONTRACTOR'S CONSTRUCTION SCHEDULE

A. Bar-Chart Schedule: Prepare a fully developed, horizontal bar-chart type Contractor's construction schedule, submit within 30 days prior to the date established for "Commencement of the Work."

B. Prepare the Schedule on a sheet to show data for the entire construction period.

C. Work stages: Indicate important stages of construction for each major portion of the work, including testing and installation.

D. Schedule Updating: Revise the schedule after each meeting or activity, where revisions have been recognized or made. Issue the updated schedule concurrently with report of each meeting.

1.05 SUBMITTAL SCHEDULE

A. Prior to beginning major construction activities, Contractor shall provide a list of submittals required for the construction of the project to include shop drawings, product data, samples, and any other information or documentation that requires review and approval by the Owner and/or Design Professional. The Contractor shall provide an anticipated delivery date next to each submittal and identify critical path items.

1.06 SHOP DRAWINGS

A. Submit newly-prepared information, on reproducible sheets, drawn to accurate scale, with the name of preparer indicated. Highlight, encircle, or otherwise indicate deviations from the Contract Documents. Do not reproduce Contract Documents or copy standard information as the basis of Shop Drawings.

B. Shop Drawings include fabrication and installation drawings, setting diagrams, schedule, patterns, templates, sectional views, complete dimensions and details, kind of materials, thickness and finish and similar drawings.

1. Sheet size: Except for templates, patterns and other full-size drawings, submit Shop Drawings in sheets at least 18" x 24", but no larger than 30" x 42".

2. Initial Submittal: Submit one correctable reproducible print and/or one digital copy in PDF format for the Design Professional’s review; the reproducible will be returned.

3. Final Submittal: Submit three hard copy prints. Two prints will be retained; the remainder will be returned.

1.07 PRODUCT DATA

A. Collect Product Data into a single submittal for each element of construction or system. Product Data includes printed information such as manufacturer's installation instructions, catalog cuts, standard color charts, and templates. Mark each copy to show applicable choices and options. Where printed Product Data includes information on several products, some of which are not required, mark copies to indicate the applicable information. For each product include; manufacturer's printed recommendations, compliance with standards application of labels and seals, notation of field measurements verified, and coordination requirements.

B. Preliminary Submittal: Submit a preliminary single copy of Product Data where selection of options is required.

C. Submittals: Submit one digital PDF copy of each required submittal. The Design Professional(s) will return the submittal marked with action taken and corrections or modifications required. Unless noncompliance with Contract Document provisions is observed, the submittal will serve as the final submittal.

1.08 SAMPLES

A. Submit full-size, fully fabricated samples cured and finished as specified and physically identical with the material or product proposed. Samples include partial sections of manufactured or fabricated components, cuts or containers of materials, color range sets, and swatches showing color, texture and pattern. Mount, display, or package samples in the manner specified to facilitate review of qualities indicated. Prepare samples to match materials as specified in the construction documents. Include information with each sample to show generic description, source, product name or manufacturer, compliance with standards, availability, and delivery time.

B. Submit samples for review of kind, color, pattern, texture for a final check of these characteristics with other elements, and for a comparison of these characteristics between the final submittal and the actual component as delivered and installed.

1. Where variation in color, pattern, texture or other characteristics are inherent in the material or product represented, submit multiple units (not less than 3), that show approximate limits of the variations.

C. Preliminary Submittals: Where Samples are for selection of color, pattern, texture or similar characteristics from a range of standard choices, submit a full set of choices for the material or product.

1. Preliminary submittals will be reviewed and returned with the Design Professional’s mark indicating selection and other action.

D. Submittal: Submit two sets; one will be returned marked with action taken. Maintain sets of Samples, as returned at the project site, for comparisons throughout the course of construction. Unless noncompliance with Contract Document provisions is observed, the submittal will serve as the final submittal.

E. Field Samples and Mock-ups specified in individual sections are special types of Samples. Field Samples are "full-size" examples erected on site to illustrate finishes, coatings, or finish materials and to establish the standard by which work will be judged.

1.09 LETTERS OF MATERIAL CERTIFICATION

A. General: Only when specified in individual sections, submit a Letter of Certification for specified materials, items, and when requested. Letters of Certification shall certify that materials submitted comply with the Contract Documents. The letter shall include substantiating supporting data, test results, and compliance with standards.

B. Submittal: Submit three copies of the Letter of Material Certification for approval.

1.10 DESIGN PROFESSIONAL'S ACTION

A. Except for submittals for record, information or similar purposes, where action and return are required or requested, the Design Professional will review each submittal, mark to indicate action taken, and return promptly. Compliance with specified characteristic is the Contractor's responsibility.

B. Action Stamp: The Design Professional will stamp each submittal with a uniform, self-explanatory action stamp. The stamp will be appropriately marked; as follows, to indicate action taken

1. Final Unrestricted Release: Where submittals are marked "Approved," that part of the work covered by the submittal may proceed provided it complies with requirements of the Contract Documents; final acceptance will depend upon that compliance.

2. Final-but-Restricted Release: Where submittals are marked "Approved as Noted," that part of the work covered by the submittal may proceed provided it complies with notations or corrections on the submittal and requirements of the Contract Documents; final acceptance will depend on that compliance.

3. Returned for Re-submittal: When submittal is marked "Not Approved, Revise and Resubmit," do not proceed with that part of the work covered by the submittal, including purchasing, fabrication, delivery, or other activity. Revise or prepare a new submittal in accordance with the notations, re-submit without delay. Repeat if necessary, to obtain a different action mark.

END SECTION 01 3300

SECTION 01 3500 - SPECIAL PROCEDURES

1.1 General: These Special Procedures are a supplement to the General Conditions of the Contract for Construction.

1.2 Drawings and Specifications: See Cover Sheet of Drawings for list of Contract Drawings. See Table of Contents of Project Specifications for list of Technical Specification Sections.

1.3 Temporary Equipment: The Contractor shall furnish, maintain and remove at completion, all equipment such as temporary ramps, chutes and like facilities, as required for proper execution of the work of all trades. The Contractor and each subcontractor shall provide, for his/her own use, all forms required for execution of his work. Such forming shall conform to requirements of authorities having jurisdiction over such work and shall be maintained in safe condition at all times and shall be removed when no longer required.

1.4 Lifting & Hoisting Devices: The Contractor shall provide, operate and maintain lifting and hoisting devices as may be required for execution of the work of all trades. Such apparatus, equipment and construction shall meet the requirements of labor laws and other applicable laws.

1.5 Temporary Support Facilities:

Sanitary Facilities: Contractor shall provide self-contained toilet units of type acceptable to the Owner, adequate for use of personnel at project site.

Water and Electric Power: The Contractor shall be responsible for obtaining or providing temporary water and electric power as necessary for construction operations. Provide temporary service, equipment, or make arrangements with the Owner for use of existing installations.

Provide service with ground-fault circuit interrupter feature activated from each circuit at a 20-amp or less rating.

The Contractor shall provide potable water adequate for personnel at project site.

The Contractor shall provide temporary security and protection. The types of provisions required include, but not by way of limitation, barricades, warning signs/lights, environmental protection, and similar provisions intended to minimize property losses, personal injuries and claims for damages at project site.

1.6 Layout of the Work: All lines, grades, levels and benchmarks shall be established and maintained by the Contractor.

Before commencing any work, the Contractor shall verify all grades, lines, levels and dimensions as indicated on the Drawings and report any errors or inconsistencies to the Design Professional(s).

The Contractor shall stake the entire project, both as to location of all construction items as well as finished grades. This stakeout may be accurate or rough, depending on the Contractor's preference. This stakeout shall be made early in the construction process and preserved as much as possible for reference during construction.

The purpose of the staking, with inspection and adjustment by the Design Professional(s), is to adapt the design to the site rather than allow the design to be forced upon the site. Staking is subject to various degrees of adaptation which can only be determined by the Design Professional(s). This variation is an aesthetic decision, and the amount of adjustment is most often determined by the existing trees, terrain, soil conditions, utilities, sub-surface water and by other intangible site conditions.

The Contractor shall notify the Design Professional(s) at least five working days before inspection of the stakeout must be made. During the inspection the Design Professional(s) will adjust the stakeout as necessary to fit the trees, topography, and all other objects and conditions on the site. This staking-inspection process must take place prior to any grading, construction, or other work on the site.

The General Contractor and superintendent shall be present during the inspection.

The staking inspection process shall be repeated for any work not staked and approved or adjusted during the first site visit. No work shall be done without the stakeout first being adjusted and approved by the Design Professional(s). All alignment, dimensions and elevation of any grading, excavation, construction, and planting is subject to adjustment to accommodate existing conditions and to save trees and other vegetation.

Any work progress delays caused by inadequate, incomplete or improper staking shall not merit an extension of the contract or delay charges by the contractor.

The Design Professional(s) shall be given at least two (2) business days to respond to any request to come to the site and adjust a stakeout.

The Design Professional(s) shall have a minimum of three (3) business days to resolve any problems created by unknown conditions discovered during the stakeout or construction.

1.7 Unknown Conditions: Subsurface Conditions: Should the Contractor encounter, during the progress of the work, subsurface latent physical conditions at the site, materially differing from those shown on the drawings or specified for unknown conditions of an unusual nature differing materially from those ordinarily encountered and generally recognized as inherent in work of the character provided for in the drawings and Specifications, the Design Professional(s) shall be notified immediately to such conditions before they are disturbed. The Design Professional(s) shall investigate the conditions. If it is found that the conditions do so materially differ, the contract price shall, with the written approval of the Owner, be

increased or decreased to accommodate with such conditions. Contract modifications will be based on unit prices as established in the Bid Form.

1.8 Existing Utilities Shown. Existing utility lines shown on the drawings, such as, cables, ducts, conduits, and piping shall, if damaged (unless they are to be abandoned) be immediately repaired, protected, and maintained in use until relocation of same has been completed or shall be cut and capped where directed or shall be prepared for service connections when so required.

1.9 Utilities Not Shown. Contractor shall be responsible for securing the services of a utility locator to determine any unknown utilities that may be on the site. Any utilities encountered that are not shown on the drawings and are to remain as active utilities, if inadvertently damaged by the Contractor, shall be repaired at the Contractor's expense. Contractor shall not proceed with any excavation until the utility locator has completed his work and reported conditions to the Owner.

1.10 Inclusion of Accessories: Unless specifically mentioned otherwise, all anchors, bolts, screws, fittings, fillers, hardware accessories, trim and other parts required for, or in connection with, an item of material to make a complete, serviceable, finished and first quality installation shall be furnished and installed as part of the item whether or not they are shown on the drawings or specified.

1.11 Protection: All materials shall be shipped, stored and handled in a manner that will afford protection and ensure their being in first class condition at the time they are incorporated in the work.

After installation all materials shall be properly protected against damage to ensure their being in first class condition when the project as a whole is completed and accepted by the Owner.

1.12 Installation: All items shall be installed in accordance with the best-recognized practices of the trade. Manufactured items shall be installed in strict accordance with the manufacturer's printed directions, specifications and/or recommendations. All working parts shall be properly adjusted after installation and left in perfect working order. Unless otherwise indicated, items exposed to weather or subject to flooding shall be installed so as to shed water. Items shall in all cases be installed plumb and true and/or in proper relation to surrounding materials.

1.13 Samples: Contractor shall be responsible for preparing samples as required in the technical specifications and to obtain approvals prior to construction of the item.

1.14 Reference to Standard Specifications: When standard specifications such as The American Society for Testing and Materials, Federal Specifications, Department of Commerce (Commercial Standards), American Institute of Steel Construction, or other well known public or trade associates are cited as a standard to govern materials, and/or workmanship, such specifications or portions thereof as referred to shall be equally as binding and have the full force and effect as though it were copied into these specifications. Such standards are generally recognized by and available to the trades concerned.

1.15 Reference to Manufacture's Publications: Unless otherwise specifically stated, all manufacturer's catalogs, specifications, instructions or other information or literature that are referred to in the specifications shall be considered as the latest edition and/or revision of such publication that is in effect on the date of the Invitation or Advertisement for Bids.

1.16 Document Signatures: Within five (5) days of notification of award or prior to execution of a contract, whichever is earliest, the Contractor shall file with the Owner a list of all persons in his firm who are authorized to sign documents such as contracts, certificates, and affidavits on behalf of the firm and to fully bind the firm to all the conditions and provisions of such documents.

1.17 Materials Furnished by Others: Whenever the Contractor or any Subcontractor shall receive items from another contractor or from the Owner for storage, erection or installation, the Contractor or Subcontractor receiving such items shall give receipts for items delivered, and any necessary replacing of item or items received. No adjustment will be made to contract price for increased insurance premiums, except for materials and/or equipment furnished by the Owner and not listed as such in other Contract Documents.

1.18 Substitute Materials and Equipment: Approval by the Design Professional(s) of substitute materials and equipment shall not relieve the Contractor from the responsibility to supply and install any additional materials, equipment, or labor required to make the substitution properly function within the intent of the Contract Documents.

1.19 Protection of Existing Structures: The Contractor shall be liable for all damage to existing structures that occurs as a result of his/her negligence to provide proper and adequate protective measures, including but not limited to buildings, walls, fences, paving, conduits, furniture, pipe, wiring, drains, underground utilities and equipment. The Contractor shall be liable for all damage to trees, shrubs, turf and other existing vegetation.

1.20 Security Considerations: Construction shall not interfere with reasonable access to and use of the adjacent facilities and structures.

1.21 Working Hours: Working hours will typically be between sunrise and sunset, Monday through Friday, excluding government holidays and any local noise ordinance restrictions. Specific hours will be established during the pre-construction conference after award of the contract.

1.22 Order of Construction: The Contractor shall submit a project schedule at the pre-construction conference outlining the anticipated construction process. Priorities within this schedule shall be coordinated with the Owner. Work is to be processed in an orderly manner. The organization of the Specifications or contract drawings does not necessarily indicate the order of sequence in which work is to be performed. Contractor shall not be granted extensions or delay charges when it is deemed that Contractor could have continued work on other components of the project or locations on the site.

1.23 Record Drawings and As-Built Documents: The Contractor shall keep a clean set of permitted drawings on-site at all times for the duration of the project. The Contractor shall keep a second set of drawings on-site and use this set of drawings to document any field changes or actual locations or dimensions that may vary from the original plans. This set will be maintained as the official Record Drawings. All notations on the Record Drawings shall be made with permanent RED ink.

Upon completion of the work, the Contractor shall mark a clean set of the appropriate contract drawings in indelible ink showing the final locations of all underground installations including, but not limited to phone lines, power lines, irrigation lines, sanitary lines, drainage lines, etc. The Contractor shall also record the proper location of all installations above ground where they have been changed on the site from designated locations on the plans. The Contractor shall provide a reproducible set of the As-Built plans to the Owner upon completion of the project.

1.24 Guarantee: The Contractor shall guarantee all work under this contract, to be free from defects of material and workmanship for a period of one year from the date of acceptance by the Design Professional(s), except as otherwise agreed upon in writing by the parties to the contract. Guarantee shall begin on the day of final acceptance, not the day of substantial completion.

Certain portions of the Work may have their own warranty and guarantee. The project warranty does not override warranties provided by product manufacturers.

1.25 Maintenance: The Contractor shall be responsible for all maintenance, as required, until completion and final acceptance of the Work. The Owner shall become responsible for maintenance upon completion and final acceptance of the work.

END OF SECTION 01 3500

SECTION 01 7700 - CLOSEOUT PROCEDURES

PART 1 - GENERAL

1.1 RELATED DOCUMENTS

A. Drawings and general provisions of the Contract and associated addenda, apply to this section.

1.2 SUMMARY

A. This Section includes administrative and procedural requirements for contract closeout, including, but not limited to, the following:

1. Inspection procedures.
2. Project Record Documents.
3. Operation and maintenance manuals.
4. Warranties.
5. Instruction of Owner's personnel.
6. Final cleaning.

1.3 SUBSTANTIAL COMPLETION

A. Preliminary Procedures: Before requesting inspection for Substantial Completion, the Contractor shall complete the following:

1. Prepare a list of items that have not yet been completed, the value of items on the list, and reasons why the Work is not complete.
2. Advise Owner of pending insurance changeover requirements.
3. Submit specific warranties, workmanship bonds, maintenance service agreements, final certifications, and similar documents.
4. Obtain and submit releases permitting Owner unrestricted use of the Work and access to services and utilities. Include occupancy permits, operating certificates, and similar releases.
5. Prepare and submit Project Record Documents, operation and maintenance manuals, damage or settlement surveys, property surveys, and similar final record information.

6. Deliver tools, spare parts, extra materials, and similar items to location designated by Owner. Label with manufacturer's name and model number where applicable.

7. Complete startup testing of systems.

8. Submit test/adjust/balance records.

9. Terminate and remove temporary facilities from Project site, along with mockups, construction tools, and similar elements.

10. Advise Owner of changeover in utilities, if applicable.

11. Submit changeover information related to Owner's occupancy, use, operation, and maintenance.

12. Complete final cleaning requirements, including touchup painting.

13. Touch up and otherwise repair and restore marred exposed finishes to eliminate visual defects.

B. Inspection:

1. Submit a written request for Substantial Completion inspection to the Landscape Architect.

2. The Design Professional(s) will inspect the work within five (5) business days of receiving the request.

3. Upon completion of the inspection, the Design Professional(s) will prepare a written report (punch list), identifying incomplete or unsatisfactory work and any other unfulfilled requirements of the contract documents. The punch list will be distributed to the Contractor and the Owner within three (3) business days of the inspection.

1.4 FINAL ACCEPTANCE

A. Preliminary Procedures: Before requesting inspection for Final Acceptance, the Contractor shall complete the following:

1. Submit a final Application for Payment.

2. Submit a final Release of Lien Waiver signed by each subcontractor who performed work on the project.

3. Submit certified copy of the punch list, stating that each item has been completed or otherwise resolved for acceptance.

4. Submit evidence of final, continuing insurance coverage complying with insurance requirements.

5. Instruct Owner's personnel in operation, adjustment, and maintenance of products, equipment, and systems.

B. Inspection:

1. Submit a written request for Final Acceptance inspection to the Design Professional(s).

2. The Design Professional(s) will inspect the work within five (5) business days of receiving the request.

3. If all work has been satisfactorily completed, the Design Professional(s) will prepare a final Certificate for Payment within three (3) business days of the inspection.

4. If any work has not been completed, the Design Professional(s) will notify the Contractor of construction that must be completed or corrected before final Certificate for Payment will be issued.

5. The Design Professional(s) shall be reimbursed by the Contractor for any time and materials associated with the re-inspection of incomplete or unsatisfactory work after the Final Acceptance inspection.

1.5 PROJECT RECORD DOCUMENTS

A. General: Do not use Project Record Documents for construction purposes. Protect Project Record Documents from deterioration and loss. Provide access to Project Record Documents for Design Professional(s) reference during normal working hours.

B. Record Drawings: Maintain and submit one set of Contract Drawings and Shop Drawings.

1. Mark Record Prints to show the actual installation where installation varies from that shown originally. Require individual or entity who obtained record data, whether individual or entity is Installer, subcontractor, or similar entity, to prepare the marked-up Record Prints.

a. Give particular attention to information on concealed elements that cannot be readily identified and recorded later.

b. Accurately record information in an understandable drawing technique.

c. Record data as soon as possible after obtaining it. Record and check the markup before enclosing concealed installations.

d. Mark Contract Drawings or Shop Drawings, whichever is most capable of showing actual physical conditions, completely and accurately. Where Shop Drawings are marked, show cross-reference on Contract Drawings.

2. Mark record sets with erasable, red-colored pencil. Use other colors to distinguish between changes for different categories of the Work at the same location.

3. Mark important additional information that was either shown schematically or omitted from original Drawings.

4. Note Construction Change Directive numbers, Change Order numbers, alternate numbers, and similar identification where applicable.

5. Identify and date each Record Drawing; include the designation "PROJECT RECORD DRAWING" in a prominent location. Organize into manageable sets; bind each set with durable paper cover sheets. Include identification on cover sheets.

C. Record Specifications: Submit one copy of Project's Specifications, including addenda and contract modifications. Mark copy to indicate the actual product installation where installation varies from that indicated in Specifications, addenda, and contract modifications.

1. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.

2. Mark copy with the proprietary name and model number of products, materials, and equipment furnished, including substitutions and product options selected.

3. Note related Change Orders, Record Drawings**,** where applicable.

D. Record Product Data: Submit one copy of each Product Data submittal. Mark one set to indicate the actual product installation where installation varies substantially from that indicated in Product Data.

1. Give particular attention to information on concealed products and installations that cannot be readily identified and recorded later.

2. Include significant changes in the product delivered to Project site and changes in manufacturer's written instructions for installation.

3. Note related Change Orders, Record Drawings**,** where applicable.

E. Miscellaneous Record Submittals: Assemble miscellaneous records required by other Specification Sections for miscellaneous record keeping and submittal in connection with actual performance of the Work. Bind or file miscellaneous records and identify each, ready for continued use and reference.

1.6 OPERATION AND MAINTENANCE MANUALS

A. Assemble a complete set of operation and maintenance data indicating the operation and maintenance of each system, subsystem, and piece of equipment not part of a system.

Include operation and maintenance data required in individual Specification Sections and as follows:

1. Operation Data:

    a. Emergency instructions and procedures.

    b. System, subsystem, and equipment descriptions, including operating standards.

    c. Operating procedures, including startup, shutdown, seasonal, and weekend operations.

    d. Description of controls and sequence of operations.

    e. Piping diagrams.

2. Maintenance Data:

    a. Manufacturer's information, including list of spare parts.

    b. Name, address, and telephone number of Installer or supplier.

    c. Maintenance procedures.

    d. Maintenance and service schedules for preventive and routine maintenance.

    e. Maintenance record forms.

    f. Sources of spare parts and maintenance materials.

    g. Copies of maintenance service agreements.

    h. Copies of warranties and bonds.

B. Organize operation and maintenance manuals into suitable sets of manageable size. Provide digital copies on flash drives in addition to paper copie. Bind and index data in heavy-duty, 3-ring, vinyl-covered, loose-leaf binders, in thickness necessary to accommodate contents, with pocket inside the covers to receive folded oversized sheets. Identify each binder on front and spine with the printed title "OPERATION AND MAINTENANCE MANUAL," Project name, and subject matter of contents.

1.7 WARRANTIES

A. Submittal Time: Submit written warranties on request of Design Professional(s) for designated portions of the Work where commencement of warranties other than date of Substantial Completion is indicated.

B. Partial Occupancy: Submit properly executed warranties within 15 days of completion of designated portions of the Work that are completed and occupied or used by Owner during construction period by separate agreement with Contractor.

C. Organize warranty documents into an orderly sequence based on the table of contents of the Project Manual. Provide digital flash drive in addition to paper copies to Owner.

1. Bind warranties and bonds in heavy-duty, 3-ring, vinyl-covered, loose-leaf binders, thickness as necessary to accommodate contents, and sized to receive 8-1/2-by-11-inch paper.

2. Provide heavy paper dividers with plastic-covered tabs for each separate warranty. Mark tab to identify the product or installation. Provide a typed description of the product or installation, including the name of the product and the name, address, and telephone number of Installer.

3. Identify each binder on the front and spine with the typed or printed title "WARRANTIES," Project name, and name of Contractor.

D. Provide additional copies of each warranty to include in operation and maintenance manuals.

PART 2 - PRODUCTS

2.1 MATERIALS

A. Cleaning Agents: Use cleaning materials and agents recommended by manufacturer or fabricator of the surface to be cleaned. Do not use cleaning agents that are potentially hazardous to health or property or that might damage finished surfaces.

PART 3 - EXECUTION

3.1 DEMONSTRATION AND TRAINING

A. Instruction: Instruct Owner's personnel to adjust, operate, and maintain systems, subsystems, and equipment not part of a system.

1. Provide instructors experienced in operation and maintenance procedures.

2. Provide instruction at mutually agreed-on times. For equipment that requires seasonal operation, provide similar instruction at the start of each season.

3. Schedule training with Owner**,** through Design Professional(s) with at least seven days advance notice.

4. Coordinate instructors, including providing notification of dates, times, length of instruction, and course content.

B. Program Structure: Develop an instruction program that includes individual training modules for each system and equipment not part of a system, as required by individual Specification Sections. For each training module, develop a learning objective and teaching outline. Include instruction for the following:

1. System design and operational philosophy.

2. Review of documentation.

3. Operations.

4. Adjustments.

5. Troubleshooting.

6. Maintenance.

7. Repair.

3.2 FINAL CLEANING

A. General: Provide final cleaning. Conduct cleaning and waste-removal operations to comply with local laws and ordinances and Federal and local environmental and antipollution regulations.

B. Cleaning: Employ experienced workers or professional cleaners for final cleaning. Clean each surface or unit to condition expected in an average commercial building cleaning and maintenance program. Comply with manufacturer's written instructions.

1. Complete the following cleaning operations before requesting inspection for Substantial Completion:

a. Clean Project site, yard, and grounds, in areas disturbed by construction activities, including landscape development areas, of rubbish, waste material, litter, and other foreign substances.

b. Sweep paved areas broom clean. Remove petrochemical spills, stains, and other foreign deposits.

c. Rake grounds that are neither planted nor paved to a smooth, even-textured surface.

d. Remove tools, construction equipment, machinery, and surplus material from Project site.

e. Remove snow and ice to provide safe access to building.

f. Clean exposed exterior hard-surfaced finishes to a dirt-free condition, free of stains, films, and similar foreign substances. Avoid disturbing natural

weathering of exterior surfaces. Restore reflective surfaces to their original condition.

g. Remove debris and surface dust from limited access spaces, including roofs, plenums, shafts, trenches, equipment vaults, manholes, attics, and similar spaces.

h. Remove labels that are not permanent.

i. Touch up and otherwise repair and restore marred, exposed finishes and surfaces. Replace finishes and surfaces that cannot be satisfactorily repaired or restored or that already show evidence of repair or restoration.

1) Do not paint over "UL" and similar labels, including mechanical and electrical nameplates.

j. Wipe surfaces of mechanical and electrical equipment, and similar equipment. Remove excess lubrication, paint and mortar droppings, and other foreign substances.

k. Replace parts subject to unusual operating conditions.

l. Clean light fixtures, lamps, globes, and reflectors to function with full efficiency. Replace burned-out bulbs, and those noticeably dimmed by hours of use, and defective and noisy starters in fluorescent and mercury vapor fixtures to comply with requirements for new fixtures.

m. Leave Project clean and ready for occupancy.

C. Comply with safety standards for cleaning. Do not burn waste materials. Do not bury debris or excess materials on Owner's property. Do not discharge volatile, harmful, or dangerous materials into drainage systems. Remove waste materials from Project site and dispose of lawfully.

END OF SECTION 01 7700